Point Set Topology on $\mathbb{R}$

A Visual Introduction to Real Analysis

Zamir Hussain
Department of Mathematics
University of Wah

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Abstract

These lecture notes provide a comprehensive, visually-driven introduction to point set topology on the real line. We systematically develop the theory of intervals, neighborhoods, and topological classifications of sets and points. The material combines rigorous mathematical definitions with intuitive geometric visualizations to build a solid foundation for advanced real analysis, functional analysis, and topology.

1 Introduction: Why Point Set Topology?

Learning Objectives

By the end of this chapter, you will be able to:

- Classify intervals and understand their topological properties
- Define and identify neighborhoods, interior points, and boundary points
- Distinguish between limit points, adherent points, and isolated points
- Characterize open and closed sets using multiple equivalent criteria
- Apply the concepts of compactness, connectedness, and density
- Compute derived sets, closures, and interiors of given sets

Chapter Overview

Chapter 1: Intervals & Neighborhoods
Chapter 2: Classification of Points
Chapter 3: Open & Closed Sets
Chapter 4: Compact & Connected Sets
Chapter 5: Advanced Topics (Perfect sets, Condensation points) **Prerequisites:**

- Real number system
- Supremum and Infimum
- Basic set theory
- Logical quantifiers ($\forall$, $\exists$)

Key Applications

Point set topology provides the foundation for:

- **Real Analysis:** Continuity, convergence, differentiation, integration
- **Functional Analysis:** Banach and Hilbert spaces, operators
- **Differential Equations:** Existence and uniqueness theorems
- **Machine Learning:** Topological data analysis, manifold learning
- **Data Science:** Cluster analysis, metric spaces, proximity measures

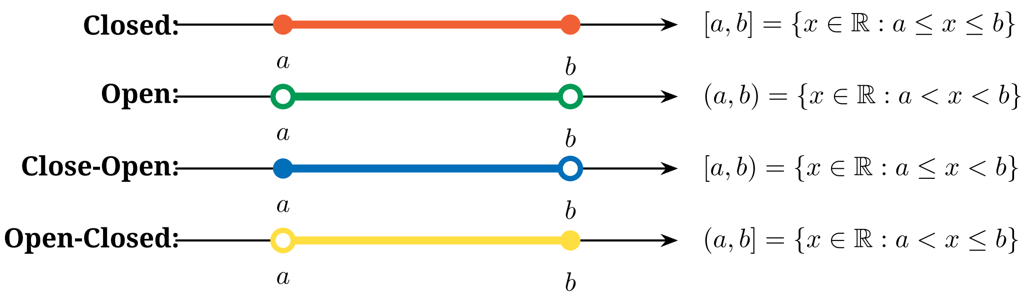
2 Intervals: The Building Blocks

Definition: Interval

An **interval** is the set of all real numbers lying between two specific real numbers a and b (where $a < b$), called the **end points**. In terms of order properties:

$$\inf(S) = a \quad \text{and} \quad \sup(S) = b$$

2.1 The Four Types of Finite Intervals



Remark

Key Properties:

- **Length:** $\ell([a, b]) = b - a = \sup(S) - \inf(S)$
- **Set Nature:** Every non-trivial interval is infinite, but not every infinite set is an interval (e.g., $(1, 2) \cup (3, 4)$)
- **Trivial Interval:** A singleton $\{a\}$ is a closed interval with zero length
- **Connectedness:** $S \subseteq \mathbb{R}$ is connected $\iff$ S is an interval

2.2 Infinite Intervals

Notation	**Set Definition**	**Type**
$(a, +\infty)$	$\{x \in \mathbb{R} : x > a\}$	Open
$[a, +\infty)$	$\{x \in \mathbb{R} : x \geq a\}$	Closed
$(-\infty, b)$	$\{x \in \mathbb{R} : x < b\}$	Open
$(-\infty, b]$	$\{x \in \mathbb{R} : x \leq b\}$	Closed
$(-\infty, +\infty)$	$\mathbb{R}$	Both open and closed

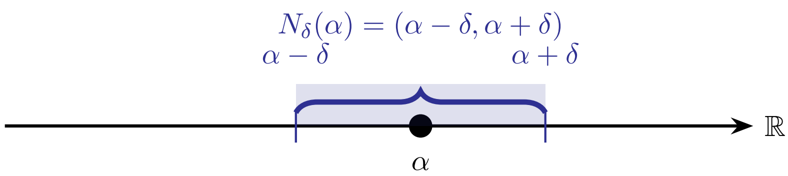
3 Neighborhoods: Local Structure

Definition: Neighborhood

Let $S \subseteq \mathbb{R}$ and $\alpha \in \mathbb{R}$. Then S is a **neighborhood** (nbd) of α if $\exists\, \delta > 0$ such that:

$$(\alpha - \delta, \alpha + \delta) \subseteq S$$

The open interval $(\alpha - \delta, \alpha + \delta)$ is called the δ-**neighborhood** of α, denoted $N_\delta(\alpha)$.



Theorem Properties of Neighborhoods

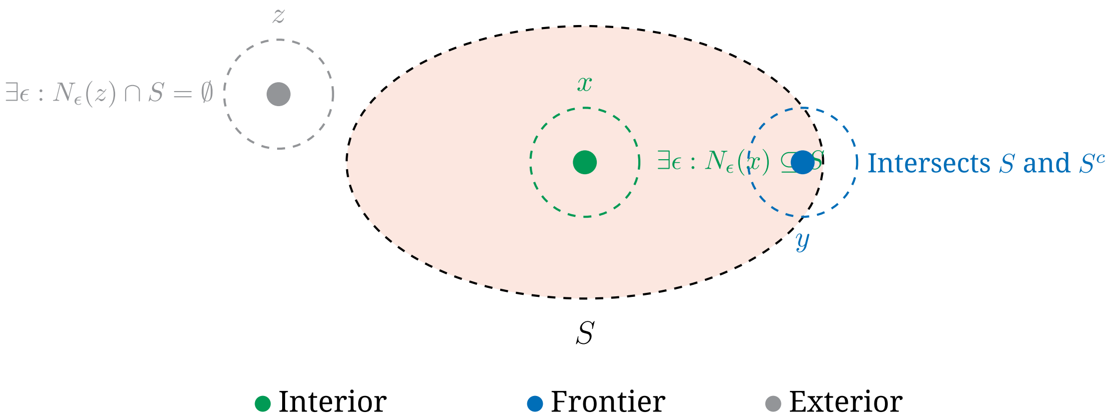
1. **Countable sets cannot be neighborhoods:** If S is countable, it cannot be a neighborhood of any point (neighborhoods contain intervals $\Rightarrow$ uncountable).
2. $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{Q}^c$ are **not** neighborhoods of any real number.
3. $\mathbb{R}$ is a neighborhood of every real number.
4. Intervals are neighborhoods of all interior points, but **not** of endpoints.
5. There are infinitely many neighborhoods for any given point.

Definition: Deleted Neighborhood

If S is a neighborhood of α, then $S \setminus \{\alpha\}$ is called a **deleted neighborhood** of α, denoted $N'_\delta(\alpha)$.

4 Classification of Points

4.1 Interior, Exterior, and Frontier Points



Definition: Interior Point

$\alpha \in S$ is an **interior point** if $\exists\, \delta > 0$ such that $(\alpha - \delta, \alpha + \delta) \subseteq S$. The set of all interior points is the **interior** S°.

Definition: Exterior Point

$\alpha \in \mathbb{R}$ is an **exterior point** of S if $\exists\, \delta > 0$ such that $(\alpha - \delta, \alpha + \delta) \cap S = \emptyset$. Equivalently, α is an interior point of S^c.

Definition: Frontier Point

$\alpha \in \mathbb{R}$ is a **frontier point** if it is neither interior nor exterior. Every neighborhood of α intersects both S and S^c. The set of frontier points is $\text{Fr}(S)$.

Example

Interior Examples:
- $[0, 1]^\circ = (0, 1)$
- $(0, 1)^\circ = (0, 1)$
- $\mathbb{N}^\circ = \emptyset$, $\mathbb{Z}^\circ = \emptyset$, $\mathbb{Q}^\circ = \emptyset$
- $\mathbb{R}^\circ = \mathbb{R}$

4.2 Boundary vs. Frontier

Definition: Boundary Point

A **boundary point** is a frontier point that belongs to S:

$$\partial S = \text{Fr}(S) \cap S$$

Set S	**Frontier Fr**(S)	**Boundary** ∂S
$[0, 1]$	$\{0, 1\}$	$\{0, 1\}$
$(0, 1)$	$\{0, 1\}$	$\emptyset$
$[0, 1)$	$\{0, 1\}$	$\{0\}$
$(0, 1]$	$\{0, 1\}$	$\{1\}$
$\mathbb{N}$	$\mathbb{N}$	$\mathbb{N}$

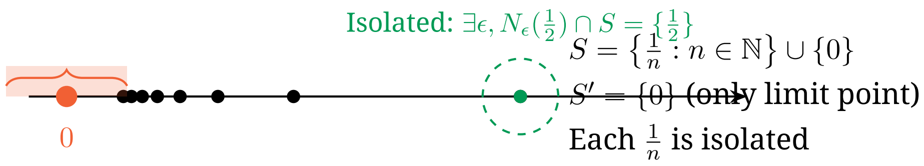
5 Limit Points and Derived Sets

Definition: Limit Point (Accumulation Point)

$\alpha \in \mathbb{R}$ is a **limit point** of S if every deleted neighborhood of α contains at least one point of S:

$$\forall \delta > 0 : \quad N'_\delta(\alpha) \cap S \neq \emptyset$$

Equivalently: Every neighborhood of α contains **infinitely many** points of S.



Definition: Derived Set

The set of all limit points of S is called the **derived set**, denoted S' or S'.

Theorem Derived Sets of Intervals

For any finite interval with endpoints a and b:

$$[a, b]' = (a, b)' = [a, b)' = (a, b]' = [a, b]$$

All four interval types have the same derived set: the closed interval $[a, b]$.

Set S	Derived Set S'
$\mathbb{R}$	$\mathbb{R}$
$\mathbb{Q}$	$\mathbb{R}$ (dense in $\mathbb{R}$)
$\mathbb{Q}^c$ (irrationals)	$\mathbb{R}$
$\mathbb{N}$, $\mathbb{Z}$	$\emptyset$ (no limit points)
$[a, b]$, (a, b), $[a, b)$, $(a, b]$	$[a, b]$
$\{1/n : n \in \mathbb{N}\}$	$\{0\}$
Finite sets	$\emptyset$

5.1 Isolated Points

Definition: Isolated Point

$\alpha \in S$ is an **isolated point** if $\exists\, \delta > 0$ such that:

$$(\alpha - \delta, \alpha + \delta) \cap S = \{\alpha\}$$

Equivalently: $\alpha \in S \setminus S'$ (belongs to S but is not a limit point).

6 Adherent Points and Closure

Definition: Adherent Point

$\alpha \in \mathbb{R}$ is an **adherent point** of S if every neighborhood of α contains at least one point of S:

$$\forall \delta > 0: \quad N_\delta(\alpha) \cap S \neq \emptyset$$

Remark

Key Distinction:

- **Limit point:** Every neighborhood contains *infinitely many* points of S (excluding possibly α itself).
- **Adherent point:** Every neighborhood contains *at least one* point of S (may be α itself if $\alpha \in S$).

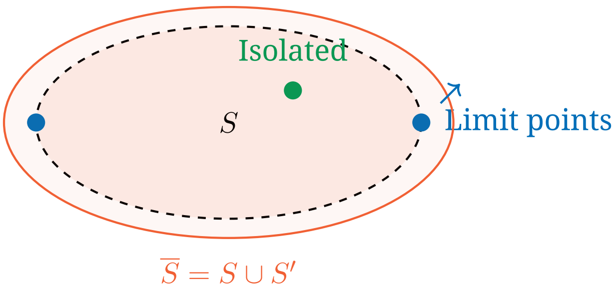
Hierarchy: Every limit point is adherent, but not conversely. Isolated points are adherent but not limit points.

Definition: Closure

The set of all adherent points of S is the **closure** of S, denoted $\overline{S}$ or $\mathrm{cl}(S)$.

Theorem Characterization of Closure

$$\overline{S} = S \cup S' = S \cup \partial S$$



Example

Closure Examples:
- $\overline{(0,1)} = [0,1]$
- $\overline{[0,1]} = [0,1]$ (already closed)
- $\overline{\mathbb{Q}} = \mathbb{R}$ (rationals are dense)
- $\overline{\{1/n : n \in \mathbb{N}\}} = \{1/n : n \in \mathbb{N}\} \cup \{0\}$
- $\overline{\mathbb{N}} = \mathbb{N}$ (no limit points)

7 Open and Closed Sets

Definition: Open Set

A set $S \subseteq \mathbb{R}$ is **open** if every point of S is an interior point:

$$S = S^\circ$$

Equivalently: S is a neighborhood of each of its points.

Theorem Properties of Open Sets

1. $\emptyset$ and $\mathbb{R}$ are open.
2. Every open interval (a, b) is an open set.
3. Arbitrary unions of open sets are open.
4. Finite intersections of open sets are open.
5. $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ are **not** open.

Definition: Closed Set

A set $S \subseteq \mathbb{R}$ is **closed** if its complement S^c is open.

Theorem Characterizations of Closed Sets

The following are equivalent:

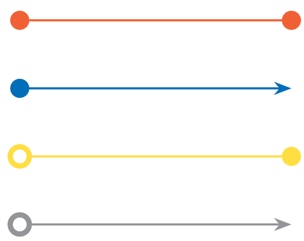
1. S is closed (complement is open).
2. S contains all its limit points: $S' \subseteq S$.
3. $S = \overline{S}$ (equals its closure).
4. Every Cauchy sequence in S converges to a point in S.

Set	**Open?**	**Closed?**
(a, b)	Yes	No
$[a, b]$	No	Yes
$[a, b)$	No	No
$(a, b]$	No	No
$\mathbb{R}$	Yes	Yes
$\emptyset$	Yes	Yes
$\mathbb{N}$, $\mathbb{Z}$	No	Yes
$\mathbb{Q}$	No	No
$\{1/n : n \in \mathbb{N}\}$	No	No

8 Compact and Connected Sets

Definition: Compact Set

A set $S \subseteq \mathbb{R}$ is **compact** if it is both **closed and bounded**.



Compact: $[0,4]$ (Closed & Bounded)

Not Compact: $[0,\infty)$ (Unbounded)

Not Compact: $(0,4]$ (Not Closed)

Not Compact: $(0,\infty)$ (Neither)

Theorem Heine-Borel Theorem

In $\mathbb{R}$, a set is compact if and only if every open cover has a finite subcover.

Definition: Connected Set

A set $S \subseteq \mathbb{R}$ is **connected** if and only if it is an interval.

Theorem

S is connected $\iff$ For all $x, y \in S$ with $x < y$, we have $[x, y] \subseteq S$.

9 Dense and Perfect Sets

Definition: Dense Set

A set $S \subseteq \mathbb{R}$ is **dense** in $\mathbb{R}$ if $\overline{S} = \mathbb{R}$, i.e., every real number is a limit point of S.

Theorem Density of Rationals and Irrationals

Both $\mathbb{Q}$ and $\mathbb{Q}^c$ are dense in $\mathbb{R}$:

$$\overline{\mathbb{Q}} = \mathbb{R} \quad \text{and} \quad \overline{\mathbb{Q}^c} = \mathbb{R}$$

Definition: Dense in Itself

A set S is **dense in itself** if every point of S is a limit point of S:

$$S \subseteq S'$$

Definition: Perfect Set

A set S is **perfect** if it is closed and dense in itself:

$$S = S'$$

Example

Examples:

- $[a, b]$ is perfect (closed, and every point is a limit point).
- (a, b) is **not** perfect (not closed).
- The Cantor set is perfect (nowhere dense, uncountable, no isolated points).
- $\{1/n : n \in \mathbb{N}\} \cup \{0\}$ is **not** perfect (0 is a limit point, but $1/n$ are isolated).

10 Condensation Points

Definition: Condensation Point

$\alpha \in \mathbb{R}$ is a **condensation point** of S if every neighborhood of α contains **uncountably many** points of S.

Adherent Points $\overline{S}$	Every neighborhood intersects S
$\supseteq$	
Limit Points S'	Every neighborhood has ∞-many points
$\supseteq$	
Condensation Points cd(S)	Every neighborhood has uncountably many

Set S	**Condensation Points cd**(S)
$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ (countable)	$\emptyset$
$\mathbb{Q}^c$ (irrationals)	$\mathbb{R}$
$\mathbb{R}$	$\mathbb{R}$
$[a, b]$, (a, b), etc.	$[a, b]$
Cantor set	Cantor set itself

11 Summary: The Hierarchy of Points

Theoretical Foundations

Complete Classification of Points Relative to a Set S

Type	Condition	Notation
Interior	$\exists\delta : N_\delta(\alpha) \subseteq S$	S°
Exterior	$\exists\delta : N_\delta(\alpha) \cap S = \emptyset$	ext(S)
Frontier	Neither interior nor exterior	Fr(S)
Boundary	Frontier $\cap\, S$	∂S
Limit	$\forall\delta : N'_\delta(\alpha) \cap S \neq \emptyset$	S'
Isolated	$\alpha \in S \setminus S'$	—
Adherent	$\forall\delta : N_\delta(\alpha) \cap S \neq \emptyset$	$\overline{S}$
Condensation	$\forall\delta : N_\delta(\alpha) \cap S$ uncountable	cd(S)

Learning Resources

Recommended Reading:

- Rudin, W. *Principles of Mathematical Analysis* (Chapters 2–3)
- Munkres, J. *Topology* (Chapter 2: Topological Spaces)
- Abbott, S. *Understanding Analysis* (Chapter 3: Topology of $\mathbb{R}$)
- Royden, H.L. *Real Analysis* (Chapter 2: Lebesgue Measure)

Online Resources:

- MIT OpenCourseWare: 18.100A Real Analysis
- 3Blue1Brown: Topology series on YouTube